

Neuromorphic Vision-Based Fall Localization in Event Streams With Temporal–Spatial Attention Weighted Network

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Abstract—Falling down is a serious problem for health and has become one of the major etiologies of accidental death for the elderly living alone. In recent years, many efforts have been paid to fall recognition based on wearable sensors or standard vision sensors. However, the prior methods have the risk of privacy leaks, and almost all these methods are based on video clips, which cannot localize where the falls occurred in long videos. For these reasons, in this article, the bioinspired vision sensor-based falls temporal localization framework is proposed. The bioinspired vision sensors, such as dynamic and active-pixel vision sensor (DAVIS) camera applied in this work responds to pixels' brightness change, and each pixel works independently and asynchronously compared to the standard vision sensors. This property makes it have a very high dynamic range and privacy preserving. First, to better represent event data, compared with the typical constant temporal window mechanism, an adaptive temporal window conversion mechanism is developed. The temporal localization framework follows a proven *proposal and classification* paradigm. Second, for the high-efficient and recall proposal generation, different from the traditional sliding window scheme, the event temporal density as the actionness score is set and the 1D-watershed algorithm to generate proposals is applied. In addition, we combine the temporal and spatial attention mechanism with our feature extraction network to temporally model the falls. Finally, to evaluate the performance of

our framework, 30 volunteers are recruited to join the simulated fall experiments. According to the results of experiments, our framework can realize precise falls temporal localization and achieve the state-of-the-art performance.

Index Terms—Bioinspired vision, falls temporal localization, privacy-preserving.

I. INTRODUCTION

AS THE WHO reported [1], the worldwide elderly population is currently rapidly increasing. The lack of nursing staff and resources has made falls a serious health care problem for the elderly living alone and the primary etiology of accidental death among people over 65 years of age [2]–[5]. There is an urgent need to develop a fall detection system for health surveillance.

In the past two decades, there has been a lot of research on fall detection, more specifically on fall recognition. Depending on the type of sensor deployed, current fall recognition systems can be broadly divided into two categories, which are based on the wearable sensor and based on the vision sensor, respectively [6]. For wearable sensor-based fall recognition systems [7]–[10], researchers typically deploy inertial measurement units, such as triaxial accelerometers, gyroscopes, and magnetometers, on different body parts of the elderly to detect abrupt changes caused by falls. However, since elderly people are required to wear specific sensors on their bodies, convenience, comfort, and accuracy are greatly limited.

Compared with wearable sensor-based fall recognition systems, noncontact vision sensor-based systems are more practical and have been used for fall recognition [11]–[14]. The traditional algorithms [11], [12] typically extract the elderly silhouette first, and then track head trajectories or body shape changes to recognize falls. Since these handcrafted feature-based algorithms are easily affected by occlusion and light changes, [13], [14] manage to apply deep learning algorithms to fall recognition. However, according to [15], the majority of falls occur between 3 A.M. and 7 A.M. The extremely low illumination condition makes it difficult for the traditional vision sensors-based camera to work properly. Furthermore, there is also risk for privacy leaks [16], as cameras need to be installed in bedrooms to monitor the elderly's movement. In addition, almost all vision sensor-based fall detection systems use video clips as input and only

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output binary classification results (fall or not), which makes it impossible to locate when a fall occurs in a long video.

To address these problems, in this article, we proposed a bioinspired vision sensor-based falls temporal localization framework. The sensor employed in this system is an event-based neuromorphic vision sensor, called the dynamic and active-pixel vision sensor (DAVIS) [17]. Compared with the traditional RGB cameras that measure the absolute brightness of all pixels at a constant frame rate, the event camera captures the brightness change of each pixel (called *event*) in real-time (microsecond level) [18]–[20]. This property makes the event-based sensors have a very high dynamic range (> 120 dB) and allows them to acquire information under extreme lighting conditions, such as dark environments. Moreover, this sparse property also makes it have the inherent ability to protect privacy.

Compared with video clip-based fall recognition, temporal localization of falls in a continuous video is a more challenging task, requiring not only recognition but also accurate localization of the temporal boundaries of the falls. In this work, a temporal-spatial attention weighted framework (TSAW) is developed to realize precisely temporal localize falls based on the DAVIS camera input event stream. The entire work adapts the proven paradigm of the *proposal and classification* and can be divided into three stages. The first stage is to generate a series of temporal range proposals that are highly relevant to the time of occurrence and duration. Unlike the traditional sliding window-based methods that generate too many redundant proposals or actionness scores-based methods that need a time-consuming actionness classifier, we propose a highly efficient event-density-based proposal strategy. After the proposal generated, the second stage is to extract temporal-spatial features through our proposed temporal-spatial attention weighted network. The attention mechanism is not novel in video understanding, and there are studies that applied the spatial attention mechanism [21], [22] or temporal attention mechanism [23] to boost the performance of action recognition. To boost falls temporal localization performance, we simultaneously combine the temporal and spatial attention mechanisms with our detection framework.

During the feature extraction, the event data representation is also a key factor. Currently, in typical computer vision fields, for example, event stream-based action or object recognition, most of the event data representation algorithms are implemented over a constant temporal window, which has an obvious disadvantage that if the motion is too fast or too slow, the represented data will be too difficult to recognize. Therefore, to achieve better event data representation, we propose to encode the event data with an adaptive temporal window.

After we obtain the extracted temporal-spatial features, a classifier and a regressor are employed to classify the candidate proposal and refine the temporal boundaries to achieve falls temporal localization.

The main contributions of this work are as follows.

- 1) To the best of our knowledge, this is the first temporal localization framework for event camera-based fall detection studies. Due to the characteristics of the event

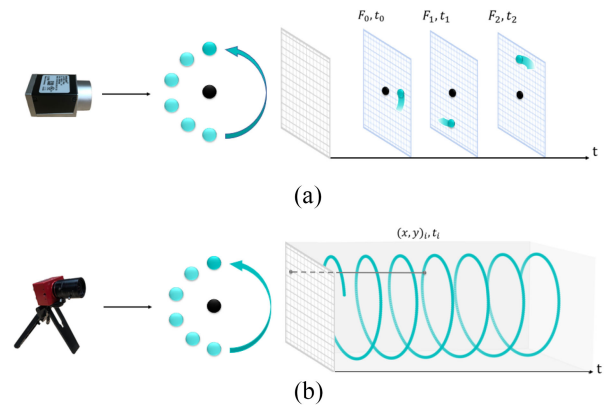


Fig. 1. Illustration of the different operating principles of (a) standard camera and (b) bioinspired event camera. In this example scene, a green ball is rotating around a static black ball. The standard camera captures both static and dynamic information at a fixed frame rate. In contrast, the bioinspired event camera only captures dynamic information and generates a series of individual events.

camera, this framework can work properly in a dark-light environment without other auxiliary devices.

- 2) We present the first event temporal density-based action proposals generation scheme for action detection, which is simpler and more effective than the existing sliding window-based schemes.
- 3) To facilitate the event camera feature representation for action recognition and detection, we develop an adaptive temporal window aggregation mechanism that can adaptively handle a variety of fast and slow actions.
- 4) To promote the development of the event-based action recognition and localization community, the dataset, related sourcecode, and experimental results of this article will be made public.¹

The remainder of this article is organized as follows. First, we review the related work in Section II, and present an overview of our TSAW fall temporal localization framework in Section III. Then, we present the experiments details, performance analyses in Section IV. Finally, we conclude and discuss our future work in Section V.

II. RELATED WORK

A. Event Camera

The event cameras are bioinspired sensors whose operating principles are completely different from the standard camera, which acquires full images at a specified frame rate (e.g., 30 fps). Event cameras respond to brightness changes in the scene *independently* and *asynchronously* for every pixel [20]. The pixel generates a new event when the incoming brightness change exceeds a specific threshold. In the static settings, since the output data generated by the event camera only contain dynamic information, the static background information is automatically filtered. Fig. 1 illustrates the operation principles differences. In the comparison scenario, the green ball periodically rotates around a static black ball. A standard camera will both capture the static and dynamic information at

¹<https://github.com/tjtum-chenlab/NeuroFall>

a fixed frame rate, while an event camera will only capture dynamic information and generate a series of individual events that represent the trajectory of the green ball. The event can be described by a quadruplet $e = (x, y, t, p)$, where (x, y) is the event's pixel location, t is the event's occurrence timestamp with a microsecond resolution, and p is the polarity representing the direction of brightness change. Moreover, the asynchronous independent response property makes the event camera have a very high dynamic range (> 120 dB), and allows it to acquire information under low lighting conditions, such as dark environments.

For event data representation, due to event camera's asynchronous sparse property, the output events are typically represented as an event set $\varepsilon = \{e_k\}_{k=1}^N$. It is fairly intuitive to consider how directly combine event set ε with a neural network to perform detection tasks. Deep spiking neural network (SNN) [24]–[26] holds great potential for directly processing these asynchronous spiking event data, and has been applied for several tasks, for example, optical-flow prediction [27]; object recognition [28], [29]; and gesture classification [30]. However, due to the lack of specified accelerating hardware and computational efficient back-propagation algorithms for asynchronous spike data, currently, the application of SNN is impractical. A typical solution is to convert these asynchronous peak data into grid-like tensors, for example, [31]–[33] that are compatible with standard convolutional neural networks that have been thoroughly researched and have specialized acceleration hardware and algorithms. In [31], in order to solve the steering angle prediction problem, the authors convert the event set ε into event count images by accumulating event data over a constant temporal window. To realize optical-flow estimation, Zhu *et al.* [32] converted events into a 4-D grid tensor that includes both polarity and normalized timestamp. Furthermore, Gehrig *et al.* [34] introduced an artificial neural network (ANN) to convert event streams into grid-like tensors intelligently.

B. Vision-Based Fall Recognition

The vision sensor-based fall recognition algorithms can be broadly divided into two categories: 1) traditional hand-crafted feature-based algorithms and 2) deep learning-based algorithms. The traditional algorithms [11], [35]–[37] typically extract the elderly silhouette first and then track head trajectories or body shape changes to recognize falls. Rougier *et al.* [11] first extracted people's silhouette and shape information then apply a gaussian mixture model (GMM) based on shape information to recognize. Li *et al.* [36] and Zhao *et al.* [37] utilized RGB-D cameras for fall recognition. The typical workflow is to use the Microsoft Kinect SDK to extract skeletal joints and human body key points information, then apply a support vector machine (SVM) to classify input RGB-D video clips.

To boost the fall recognition performance, Lu *et al.* [13], Zhang *et al.* [14], and Lee *et al.* [38] introduced deep learning for fall recognition. In [13], they use 3-D CNN kernels to encode spatial and short temporal range information over video clips and apply the LSTM network to combine the extracted

long temporal range features. However, 3-D convolution consumes huge computational resources, Lee *et al.* [38] utilized a temporal segment network proposed in [39] and realized a real-time recognition speed on an embedded device.

C. Temporal Action Detection

Based on the detection strategy, temporal action detection algorithms can be divided into three categories: 1) methods perform frame-level classification and apply smoothing and merging algorithms to obtain temporal boundaries [40], [41]; 2) methods adapt a two-stage *proposal and classification* paradigm involving proposal generation [42]–[44], classification, and temporal boundary refinements [43], [45]; and 3) methods develop an end-to-end framework, which integrates the proposal generation and recognition [46], [47]. In this work, we adapt the proven paradigm of proposal and classification and propose our TSAW.

1) *Temporal Proposal Generation*: For the proposal generation, currently, there are two main categories of methods: 1) sliding window-based methods [48], [49] and 2) actionness score-based methods [43], [44]. The sliding-window-based method typically scans a continuous video stream at fixed intervals and generates temporal range proposals of various predefined scales. Although this sliding window-based method has achieved great success in a general object detection work [50] and is intuitive and straightforward. It is not suitable for temporary action detection, especially for sparse action detection (such as falls), because too many redundant proposals require substantial computing resources to train and inference, which severely limits its practical application. The actionness score-based methods can generate more accurate and less redundant proposals. This method usually finds those continuous temporal regions with mostly high actionness scores to serve as proposals. The actionness score was first introduced in [51] for spatial action detection and then introduced into the temporal proposal generation for action localization. Zhao *et al.* [43] devised an effective proposal method called TAG, which uses a classic watershed algorithm [52] applying with 1-D actionness scores. Unlike Zhao *et al.* [43] watershed algorithm-based action proposal generation, Lin *et al.* [44] devised a boundary sensitive network for temporal action proposal generation called BSN. They evaluate the boundaries and actionness probabilities of each temporal location and then generate temporal proposals based on boundary permutation. Yet, all the above action proposal generation methods are either based on sliding-window that generate too many redundant proposals or based on actionness scores that need a pretrained binary actionness classifier network which can also be very time consuming. Due to the event camera asynchronous sparseness and motion-sensitive properties, we directly set event density as actionness scores to mine interest proposals.

2) *Temporal Feature Modeling*: For temporal feature modeling, currently, there are also three main categories temporal modeling strategies for general video clips. The intuitive and straightforward method is to apply a 3-D convolutional neural network to jointly learn spatial-temporal features based

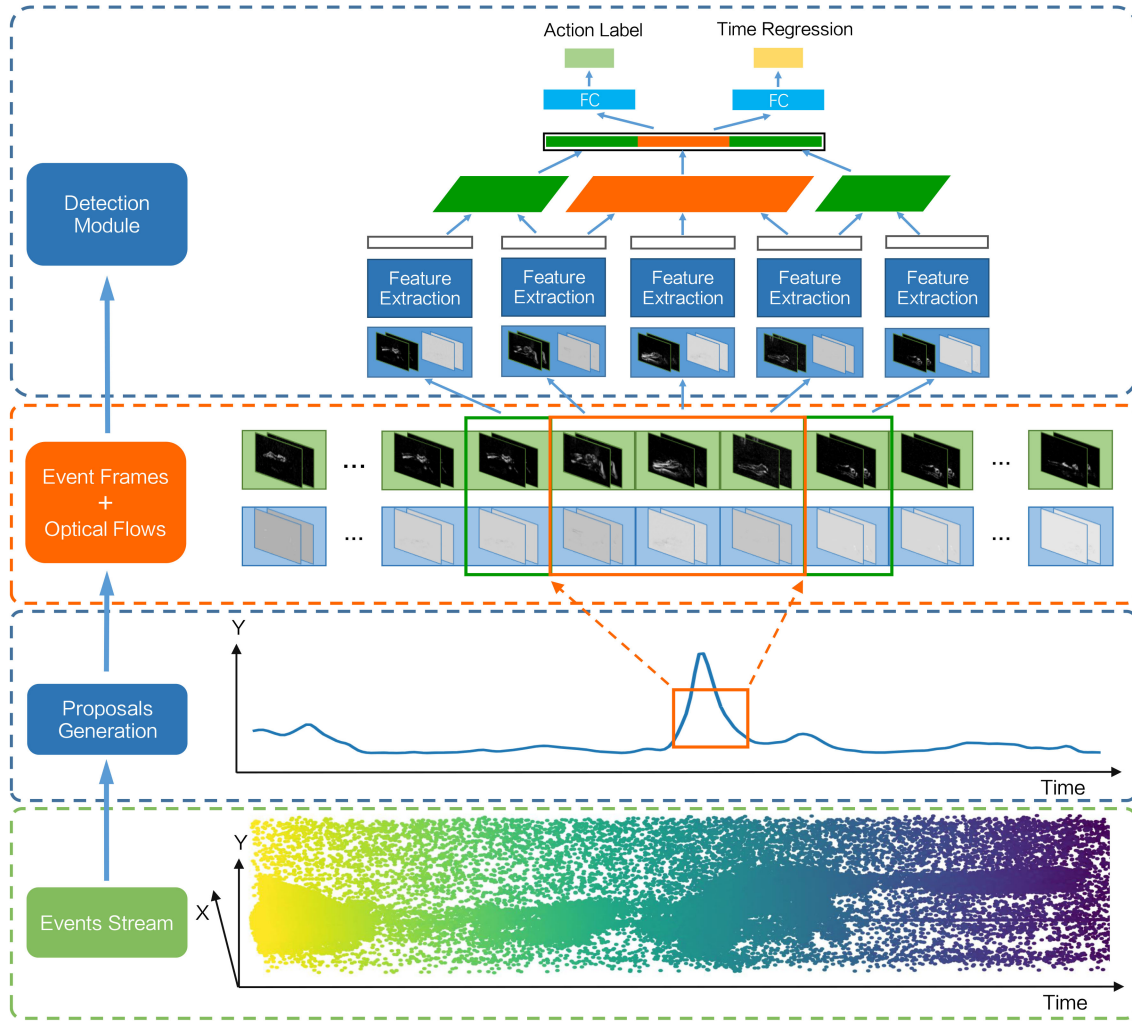


Fig. 2. Overview of our proposed event-camera-based falling down action detection framework. On each input events-stream, we first use our temporal proposals generation module to produce highly relevant candidates (orange boxes). After that, we expand each candidate proposal toward forward and backward (green box) to capture how actions start, proceed, and end. Based on the augmented candidate temporal proposal, we apply our feature extraction backbone network to extract temporal-spatial weighted proposal features. Thereafter, a proposal classifier and temporal boundary regressor are introduced to classify the proposal and refine the proposal's temporal boundary.

on continuous frames inputs [53]–[56]. However, consecutive frames contain too much redundant information, which makes 3-D CNN networks computationally expensive and difficult to deploy. A better method is to sparsely sample frames with a predefined interval τ ($\tau \geq 2$), that is, sampling $k, k + \tau, k + 2\tau, \dots, k + n\tau$ frames. Though this sparse sampling strategy is more advisable than the consecutive, it is still a suboptimal method, and its performance is directly related to the sampling interval. With τ too small, this sparse sampling method degenerates to the consecutive sampling; with τ too large, some key frame information may be lost. The optimal temporal modeling is to divide the video clip into K segments $V = \{S_1, S_2, \dots, S_K\}$ of equal duration [39], and for each segment $S_i \in V$ randomly sampling key feature frames.

The attention mechanism is not novel in video understanding. Sharma *et al.* [21] took RGB frames as input and utilized CNN models as backbones to extract spatial attention weighted features for general action recognition. However, this unconstrained spatial soft attention is apt to be distracted by static background features. To extract more robust motion features,

Peng and Schmid [57] and Wang *et al.* [58] used the state-of-the-art object detection networks to extract region of interests and adopt RoI pooling strategy to extract more informative data. Besides the spatial attention mechanism, Zang *et al.* [23] also introduced the temporal attention mechanism to action recognition.

In this work, to achieve effective and precise falls temporal detection, we utilize the third temporal segment based on proposal modeling scheme and combine with the temporal and spatial attention mechanism simultaneously.

III. METHODOLOGY

A. Overview

Fig. 2 illustrates our proposed event-camera-based fall localization framework, which takes events stream as inputs and outputs predicted falls' categories and temporal ranges (represented by a start and end boundary). The entire falls localization system can be divided into three stages. The first stage is to produce a set of *temporal range proposals* that

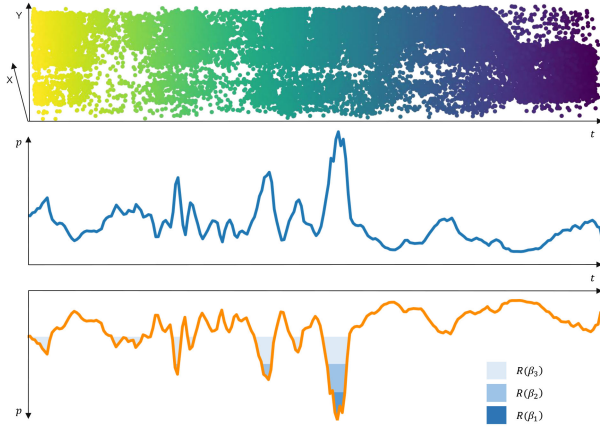


Fig. 3. Visualization of our event density-based temporal proposals generation method. The top picture visualizes an input event stream in points at $t-x-y$ 3-D coordinates. The middle picture is the corresponding actionness score curve of the input event stream based on the events-temporal density. The bottom picture presents the watershed algorithm-based temporal proposals generation at different water-level β_i , and the corresponding proposed temporal proposal regions $R(\beta_i)$.

are highly relevant to the falls' occurrence and duration. The temporal proposal generation methods will be discussed comprehensively in Section III-B. For each temporal proposals, our fall localization framework will extract temporal and spatial attention weighted features with our feature extraction backbone network. The proposal feature extraction details will be discussed in Section III-D. Finally, we introduce a classifier and a temporal boundary regressor to classify the candidate proposal and refine the temporal boundaries, respectively.

B. Proposal Generation

Since an effective proposal generating method that produces more accurate proposals is of vital importance for action detection network to achieve better performance. In this article, based on the DAVIS camera's asynchronous sparseness and motion-sensitive properties, we propose a highly efficient event-based action proposal strategy that sets events temporal density as actionness scores. For an arbitrary input events-stream, we first sample the actionness scores at regular intervals of t_i . To this end, we can divide the input events-stream into segments $\mathcal{S}$ at the fixed sampling intervals of t_i . For a specific segment $s_i \in \mathcal{S}$, we count the events amount as $E_c(s_i)$, and the actionness score $P(s_i)$ of this segment can be calculated as

$$P(s_i) = \frac{E_c(s_i) - \min\{E_c(s_k)\}}{\max\{E_c(s_k)\} - \min\{E_c(s_k)\}}, \quad s_k \in \mathcal{S}. \quad (1)$$

After obtaining the input event stream's actionness scores, we then apply the classic watershed algorithm to generate temporal action proposals. The watershed algorithm is a well-studied method in mathematical morphology and has been applied to image segmentation since the 1970s [59], [60]. The basic idea is fairly intuitive. We can imagine the event stream's actionness scores as a 1-D terrain and then submerge this terrain from its minima to a specific water-level β to obtain different submerged temporal regions $R(\beta)$. Each $r_i \in R(\beta)$ corresponds to a temporal region that has actionness score $p \geq \beta$. Fig. 3 illustrates the watershed algorithm-based proposal generation scheme.

According to our extensive experiments, we sample the water-level $\beta \in (0, 1)$ with an even step size of 0.1 and obtain the entire temporal proposals as R . We then apply nonmaximum suppression to proposals with an IoU threshold of 0.90 to filter out highly overlapped proposals. In addition, we also introduce a group strategy employed in [43] to concatenate proposals at different temporal IoUs.

C. Event Data Representation

As we mentioned before, the bioinspired camera's output events represented as an event set $\varepsilon = \{e_k\}_{k=i}^N$ are typically converted to grid-like tensors, which make them compatible with standard convolutional neural networks. However, most existing conversion schemes are implemented over a constant temporal window, which has an obvious disadvantage that if the motion is too fast, many asynchronous events will be accumulated, which will make the converted grid tensor too blurry to recognize. If the motion is too slow, only a few asynchronous events will be accumulated, which will result in a low signal-to-noise ratio and converted grid tensor too barren to recognize.

To achieve a better event data representation, we convert the event set ε into grid tensors with an adaptive temporal window based on the accumulated event data amount. As long as the amount of event data exceeds a predefined threshold, we rotate the temporal-window to start the next conversion. We compare the converted grid-like tensor on slow motion and fast motion scenes by an adaptive temporal window and a constant temporal window in Fig. 4, respectively. Both of slow motion and fast motion scenes share the same converting parameter. For the constant temporal-window-based conversion method, the temporal window is set as 0.05 s. As for the adaptive temporal window-based conversion method, through some comparative experiments, the predefined event amount threshold is set to 1000.

From Fig. 4, we can infer that the adaptive temporal window-based event data conversion method can realize more robust data representation compared to the constant temporal window-based methods on both slow motion and fast scenes.

For event data in a specified conversion temporal window, two channels are utilized to represent the spatial and temporal information. The first channel, we follow [61] to use event occurrence frequency at a specific pixel location to encode the converted pixel's value, and the value σ can be calculated as:

$$\sigma(x, y) = 255 \times \frac{1 - e^{E_c(x, y)}}{1 + e^{E_c(x, y)}} \quad (2)$$

where x, y is the pixel's location, and $E_c(x, y)$ is the event occurrence frequency at pixel (x, y) . With this frequency encoding method, the noise data can be significantly suppressed.

For the second channel, we use [62] proposed surface of active events (SAEs) encoding method. For each pixel, the value $g(x, y)$ is set by the most recent timestamp $t_{x, y}$ as

$$g(x, y) = 255 \times \frac{t_{x, y} - t_{\text{start}}}{t_{\text{end}} - t_{\text{start}}} \quad (3)$$

where t_{start} and t_{end} are current adaptive temporal window's start and end tick. Although this encoding method highly suppresses spatial texture information, the gradient of pixels

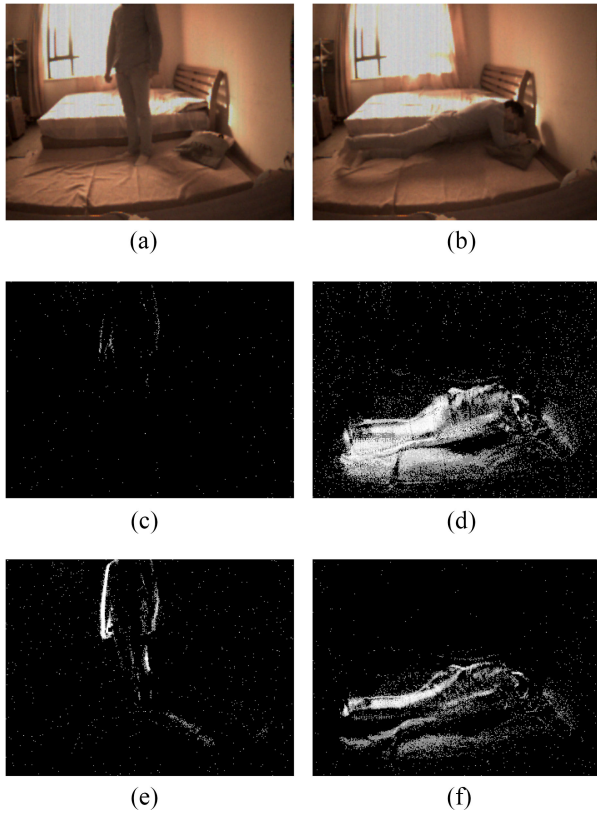


Fig. 4. Comparison between grid-like event tensors transformed by an adaptive temporal window and a constant temporal window on slow motion and fast motion scenes. To make clarity, we also present the simultaneously recorded RGB images [(a) and (b)] by the DAVIS camera under the converting scenarios. (e) and (f) are converted by an adaptive temporal window. (c) and (d) are converted by a constant temporal window.

values can tell the speed and direction of motion, which can be complementary to our first channel's information.

D. Temporal-Spatial Attention Weighted Network

1) *Temporal Modeling*: For an input event stream, the event temporal density-based proposal generation module will produce a set of N proposals $P = \{p_i = [t_{si}, t_{ei}]\}_i^N$. Since each proposal is generated at high event-temporal density, in order to capture how action start and ends, we augment proposal $p_i = [t_{si}, t_{ei}]$ to $p'_i = [t'_{si}, t'_{ei}]$, where $t'_{si} = \max\{0, t_{si} - d_i/2\}$, $t'_{ei} = \min\{T_d, t_{ei} + d_i/2\}$, T_d is the input event stream's duration, and $d_i = t_{ei} - t_{si}$ is proposal p_i 's temporal range. The two augmented part $p'_i = [t'_{si}, t_{si}]$ and $p'_i = [t_{ei}, t'_{ei}]$ are corresponding to the proposal's *starting* and *ending* stages, respectively.

As we mentioned before, we use the temporal segment sparse sampling strategy to model each proposal. Inspired by [39], [63], and [64] that the combination of multi modality data can be beneficial for spatial and temporal motion information representation, in this work, for event data-based temporal action modeling, we combine the event optical flow and event converted two-channel grid tensor as input data. For the asynchronous sparse event set's optical-flow estimation, we directly apply the adaptive block-matching optical-flow (ABMOF) algorithm proposed in our previous work [65].

2) *Temporal-Spatial Attention Mechanism*: Our proposed TSAW network is presented in Fig. 5. Given proposal $p_i = [t_{si}, t_{ei}]$ and the temporal augmented part $p'_i = [t'_{si}, t_{si}]$ and $p'_i = [t_{ei}, t'_{ei}]$, we divide it into N_c , N_s , and N_e segments, respectively. For each segment S_i , we randomly sample event optical flow and event converted two-channel tensor in segment's coverage as input. Then, the temporal attention CNN stream maps the optical-flow input to spatial attention vector w_{S_i} . The spatial CNN stream maps the event converted two-channel tensor to spatial feature vector f_{S_i} . Since optical-flow data encode motion information, to extract more valuable motion features and suppress useless static information, we use optical-flow stream output spatial attention vector w_{S_i} to multiply the spatial feature vector f_{S_i} to obtain the spatial weighted feature vector f'_{S_i}

$$f'_{S_i} = w_{S_i} \odot f_{S_i}. \quad (4)$$

In addition to the spatial attention vector w_{S_i} , for each segment S_i , we also employ a multilayer perceptron to obtain current segment's temporal attention weight w_{ti} used to decide which segments should we pay more attention to. w_{ti} is formulated as

$$e_{ti} = f_{nn}(w_{S_i}) \quad (5)$$

$$w_{ti} = \frac{\exp(e_{ti})}{\sum_{j=0}^N \exp(e_{tj})}, \quad N = \{N_s, N_c, N_e\} \quad (6)$$

where f_{nn} is the multilayer perceptron model and w_{S_i} is segment's spatial attention vector. We use (6) to calculate normalized temporal weight through a soft-max function. For each stage, we have $\sum_N w_{tj} = 1$, where N is the current stage's segment amount.

Based on the calculated temporal weight w_{ti} , the stagewise temporal weighted pooling is formulated as

$$f_{att} = \sum_{i=0}^N w_{ti} \cdot f'_{S_i}, \quad N = \{N_s, N_c, N_e\} \quad (7)$$

where f'_{S_i} is the spatial-attention weighted feature vector and f_{att} is the current stage's temporal attention weighted feature vector. Clearly, if $w_{ti} \equiv (1/N)$, every segment's importance is equal, the temporal weighted pooling is degenerated to an average pooling. After obtaining the stagewise feature, we combine the three-stage features into an overall feature to represent the proposal's temporal-spatial characteristics.

3) *Classification and Localization Regression*: To predict each proposal's action label, based on the temporal weighted high level feature, we introduce a multilayer perceptron-based classifier to classify input proposal into $K + 1$ classes, that is, one *background* class with label 0 and K action classes with label 1, $\dots$, K . The output of the classifier is normalized by a soft-max layer, and the value can be considered as a corresponding action class probability. From a probabilistic view, the value can be considered as a conditional distribution $P(c_i|p_i)$, where c_i is the class label and p_i is current input proposal.

In addition to proposal classification, we also introduce a *temporal boundary regression* module to refine proposal's temporal range. This module is also constructed based on a multilayer perceptron and output a temporal range refinement

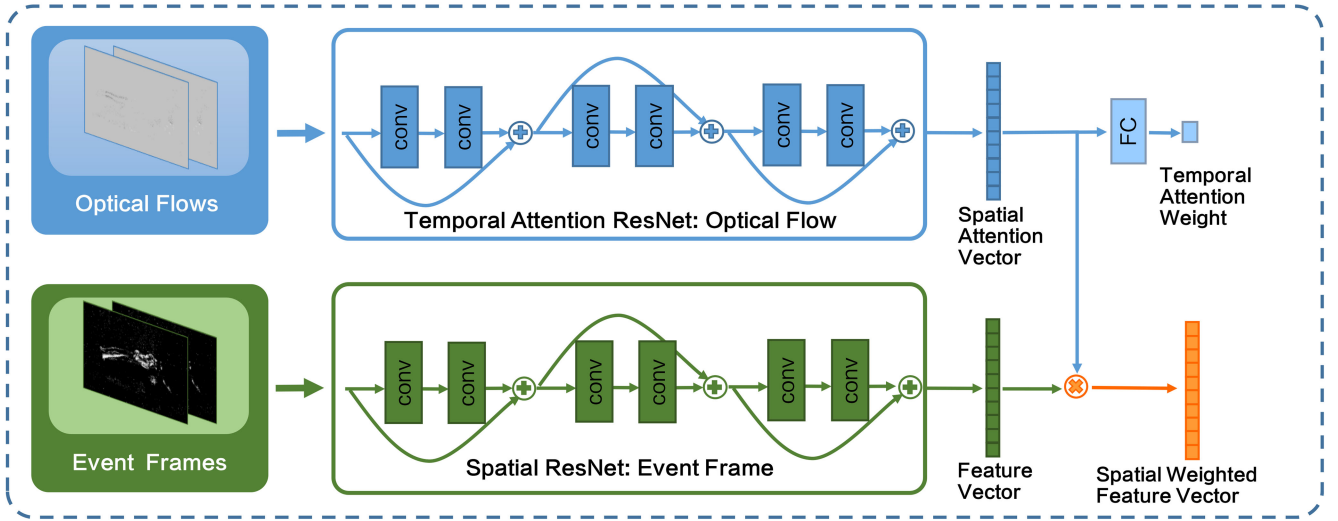


Fig. 5. Proposed TSAW architecture. For each segment, there are two stream inputs, optical flows and event converted two-channel tensors. The optical-flow stream is used to obtain the temporal and spatial attention weight of the current segment. The event tensor stream is used to extract temporal and spatial weighted motion features of current segment.

for each action class. We follow the regression target design of SSN [43], that is, for each positive proposal p_i , we regress $R = \{r_k = [u_k, \phi_k]\}_{k=1}^K$ where K is the action class amount, u_k is temporal center relative shift for the k th class action, and ϕ_k is temporal range relative shift for the k th class. For each positive proposal, we use the closest ground-truth instances as the regression target.

To simultaneously train the classification module and temporal boundary regression module, we define a multitask training loss as

$$\mathcal{L}_{cls} = - \sum_{k=0}^K y_k \cdot 1_{(c_i=k)} \log P(c_i|p_i) \quad (8)$$

$$\mathcal{L}_{reg} = -\text{SmoothL1}(u_i, \phi_i, p_i) \quad (9)$$

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \cdot 1_{(c_i \geq 1)} \mathcal{L}_{reg}. \quad (10)$$

We use [66] proposed smooth L1 loss function for $\mathcal{L}_{reg}$ because it prevents gradient from exploding when regressions are too far away from targets, and also keep gradient sensitive when regressions are close to the optimum targets.

IV. EXPERIMENTAL RESULTS

We conducted experiments on event-camera falls datasets to evaluate our proposed framework's performance. In this section, we will first discuss our recorded datasets and the implementation details. Then, we will investigate the impact of some different settings on our framework via a set of ablation studies. Finally, we will compare our proposed framework's performance with other state-of-the-art approaches.

A. Dataset

To the best of our knowledge, different from traditional cameras, there is currently no public event-camera-based activity temporal localization datasets. The already opensource event-camera-based datasets are focused more on optical-flow

estimation [67], action recognition [68], [69], or visual odometry [70].

For these reasons, we recruited 30 participants to record 900 long-time event-streams (around 60 s) using event camera.² 60% of them were recorded under good lighting conditions, and the rest were recorded under dark lighting conditions. Thereafter, we divide them into about 4500 short event streams (around 10 s). All of them involve normal actions like sitting in the chair, lying on the bed, walking around and fall-like actions like bending or kneeling on the floor. 40% of short event streams contains falling actions (falling from a bed or chair). In Figs. 6 and 7, some recorded example actions under good or dark illumination condition are presented, respectively. From Fig. 7, it can be inferred that in dark lighting conditions, the event camera can still work properly while the traditional RGB vision cameras are almost nonfunctional.

B. Implementation Details

For more efficiently extracting falling down activities' spatiotemporal features and to avoid training difficulties caused by sample imbalance, during training, for each minibatch, we specified the positive proposals and negative proposals amount. For example, the sample amount for positive and negative proposals are set to 3 and 5, respectively. As we mentioned before, given a proposal $p_i = [t_{si}, t_{ei}]$ and the temporal augmented parts $p_i^s = [t'_{si}, t_{si}]$ and $p_i^e = [t_{ei}, t'_{ei}]$, we divide it into N_c, N_s, N_e segments, respectively. In our experiments, we set $N_c = 3, N_s = N_e = 1$. For temporal-spatial feature extraction, we utilize two streams like network [64]. For each segment optical-flow input, we sample two consecutive optical-flow frames containing horizontal and vertical two channels. As for event tensor input, we also sample two consecutive event tensors that contain SAE and Frequency channels. Our TSAW feature extraction network is constructed based on ResNet [71] which are pretrained on ImageNet [72].

²Once this article is accepted, our dataset will be made public.

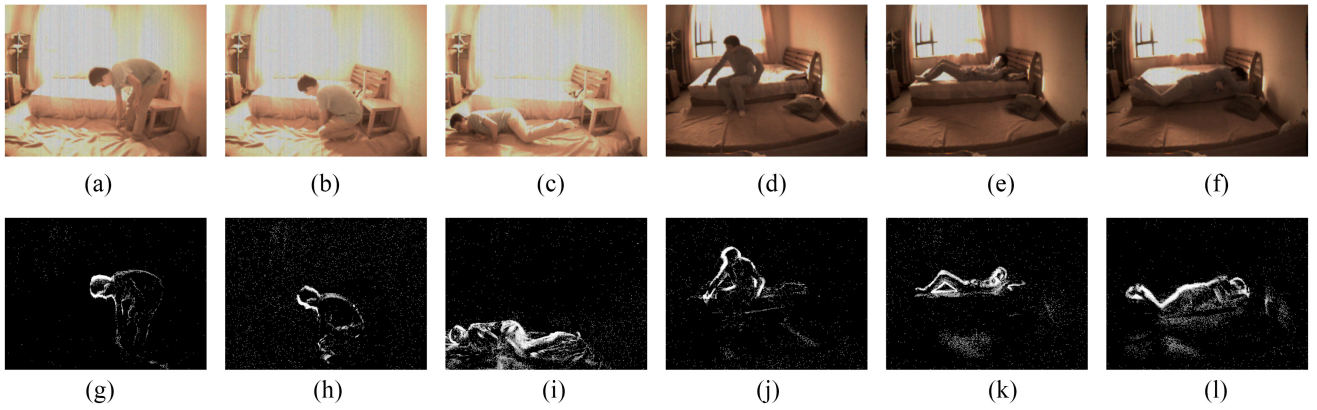


Fig. 6. Examples of our recorded event camera-based fall temporal detection dataset under good illumination. (a) to (f) are DAVIS camera simultaneously recorded RGB images, and the (g) to (l) are the corresponding event stream tensor images converted with an adaptive temporal window mechanism introduced in Section III-C.

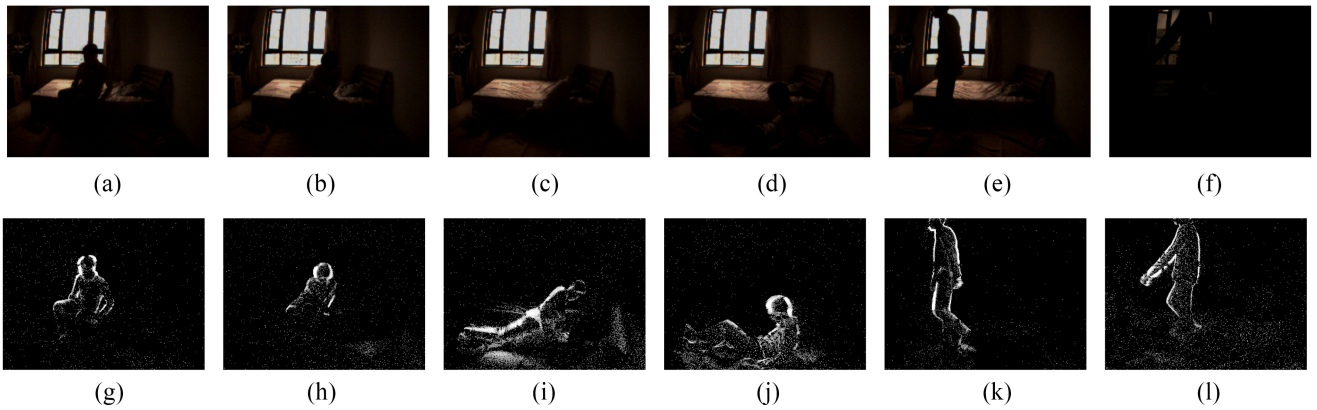


Fig. 7. Examples of our recorded event camera-based fall temporal detection dataset under bad illumination. (a) to (f) are DAVIS camera simultaneously recorded RGB images, and the (g) to (l) are the corresponding event stream tensor images converted with an adaptive temporal window mechanism introduced in Section III-C.

The difference is that we replace the first convolutional layer to fit our input channels for optical flow and event-grid tensors. For extracting more accurate spatial attention vector, during the data augmentation process, we bundled optical flow and event-grid tensors and rescaled the minimum spatial dimension to 256 and then performed a size of 224×224 random crop.

We split our dataset into three parts: 1) *train*; 2) *validation*; and 3) *test* with a ratio of 7 : 1 : 2. Our network is implemented with PyTorch-1.4 on an RTX-2080 Ti GPU. During training, we utilize an Adam optimizer with learning rate of $1e-4$ to optimize our framework's parameters. The hyperparameter λ is set to 0.1. For reporting our framework's performance, we use the mean average precision (mAP) at different temporal IoU thresholds as evaluation metric. The temporal IoU thresholds are set to $\{0.1, 0.3, 0.5, 0.7\}$.

C. Ablation Studies

1) *Proposal Generation*: In addition to the event temporal density-based proposal scheme, we also introduce two more specific optical-flow-based proposal schemes for falls localization. Compared to normal activities, the falls involve much more vertical downward movement, that is, the value of

TABLE I
COMPARISON OF DIFFERENT PROPOSAL METHODS' PERFORMANCE WITH THE SAME NUMBER OF PROPOSALS. WE USE AVERAGE RECALL RATES (AR) AS EVALUATION METRIC

Proposal Method	AR(%)		
	Top 20	Top 30	Top 50
Sliding Window	6.76	9.12	14.21
Optical Flow Downward Speed Based	43.56	45.62	46.34
Optical Flow Downward Density Based	47.81	55.45	63.10
Event Temporal Density Based	60.19	64.47	66.23

downward optical-flow density or speed may also be used as actionness scores for falls proposals. Different from the traditional frame-based optical-flow estimation, our previously proposed algorithm ABMOF [65] can realize asynchronous sparse point optical-flow estimation. With this algorithm, we can easily obtain downward optical-flow density or speed of input event stream. For comparison, the sliding window proposal scheme is also introduced. Since the duration of almost all falls is short, the window size is set as $\{0.3 \text{ s}, 0.5 \text{ s}, 0.7 \text{ s}\}$ and step is set as 0.1 s.

We evaluate the average recall rates (AR) for each proposal scheme, and the results are summarized in Table I and plot

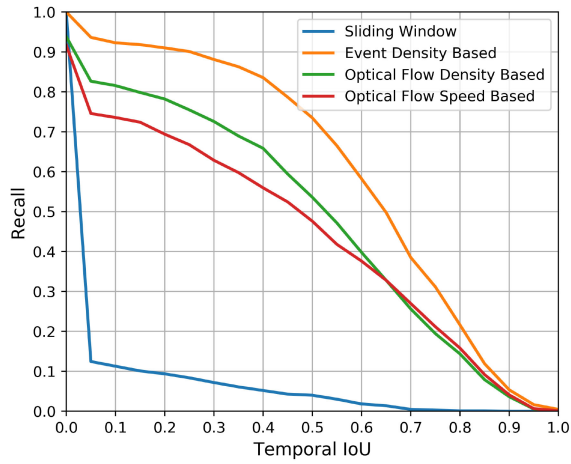


Fig. 8. Recall rate of the top 50 proposals with different proposal methods at different temporal IoU.

TABLE II

COMPARISON OF DIFFERENT EVENT DATA REPRESENTATION METHOD'S PERFORMANCE WITH THE SAME FRAMEWORK CONFIGURES. WE USE MAP AS EVALUATION METRIC

	mAP(%)@IoU				
	0.1	0.3	0.5	0.7	Average
Constant Time based	84.50	83.68	81.79	74.38	81.09
Adaptive Time based	88.62	88.03	86.24	78.75	85.41

the AR curve in Fig. 8. We can see that the *event temporal density-based* proposal scheme has higher AR at the same proposals amount. In particular, compared to the sliding window proposal scheme, the event temporal density based has an AR gain of more than 50% at the top 20 proposals, which indicates that the event temporal density-based scheme is a more efficient and can generate more accurate proposals. The two optical-flow-based methods also have competitive performance compared to the sliding window scheme. The main bottleneck of these methods may be the performance of optical-flow algorithms.

2) *Event Data Representation*: To measure how much performance improvement the adaptive temporal window-based event data representation method can bring compared to the constant temporal window, we evaluate the mAP performance of these two data presentation methods, and the results are summarized in Table II. We can see that under the same network configuration conditions, the adaptive temporal window-based event data representation can bring about an absolute 4% performance improvement. We attribute the performance gain to the adaptive temporal window-based event data representation method that captures more detailed, robust features regardless of motions speed.

3) *Influence of Attention Modules*: Here, to evaluate the necessities of the temporal attention module and the spatial attention module, several ablation experiments are conducted on our proposed TSAW framework. The results are summarized in Table III. As expected, the introduction of both the spatial attention module and the temporal attention module can improve the performance of our framework. Specifically,

TABLE III

INFLUENCE STUDY OF ATTENTION MODULES, WHERE TA REPRESENTS THE TEMPORAL ATTENTION MODULE AND SA REPRESENTS THE SPATIAL ATTENTION MODULE. IN THE EXPERIMENTS, EXCEPT FOR SUBTRACTING SPECIFIC MODULES, THE OTHER FRAMEWORK CONFIGURATIONS ARE THE SAME

Framework Configure	mAP(%)@IoU				
	0.1	0.3	0.5	0.7	Average
Without TA	87.60	86.92	84.07	76.13	83.68
Without SA	87.29	86.49	84.71	76.69	83.80
Without TA and SA	87.55	86.00	81.82	72.60	81.99
Full	88.62	88.03	86.24	78.75	85.41

TABLE IV

EVENT STREAM-BASED FALLING DOWN ACTION DETECTION RESULTS, MEASURED BY MAP AT DIFFERENT TEMPORAL IOU THRESHOLDS. TA REPRESENTS THE TEMPORAL ATTENTION MODULE, AND SA REPRESENTS THE SPATIAL ATTENTION MODULE

Method	mAP(%)@IoU				
	0.1	0.3	0.5	0.7	Average
SSN [43]	87.55	87.12	83.93	74.01	83.15
TSAW without TA	87.60	86.92	84.07	76.13	83.68
TSAW without SA	87.29	86.49	84.71	76.69	83.80
TSAW	88.62	88.03	86.24	78.75	85.41

for the spatial attention module, when IoU is set to 0.7, it can help to obtain an mAP performance gain of about 4% compared without the spatial attention module. We attribute this performance gain to the module suppressing static functions and capturing more accurate motion patterns of different actions. For the temporal attention module, it also helps to obtain a notable performance gain compared with no temporal attention module models. The reason is that different actions have different stage characteristics. Through the temporal attention module, our framework can learn to pay attention to the more representative stage characteristics of different actions. For these reasons, we combine the spatial attention module and the temporal attention module to the detection framework and form the TSAW. Through Table III, we can see that the combination of the spatial attention module and the temporal attention module can help to obtain about 6% mAP performance gain at IoU = 0.7, and about 4% average mAP performance gain at different IoU thresholds.

D. Comparison With State of the Art

Finally, we compare our TSAW action temporal localization framework based on event stream with the other state-of-the-art methods and analyze the performance with the same evaluation metric used before. The experiment results are summarized in Table IV. For SSN [43], we follow the authors' setting and train on our dataset from scratch. According to the experimental results, our proposed TSAW framework has better performance than SSN. Specifically, when IoU is set to 0.7, the TSAW framework has about 4% performance gain than SSN, and about average 2.5% performance gain at different IoU thresholds. Even if one of the attention modules is removed, our method still has a competitive performance compared to SSN. the qualitative results are visualized on the

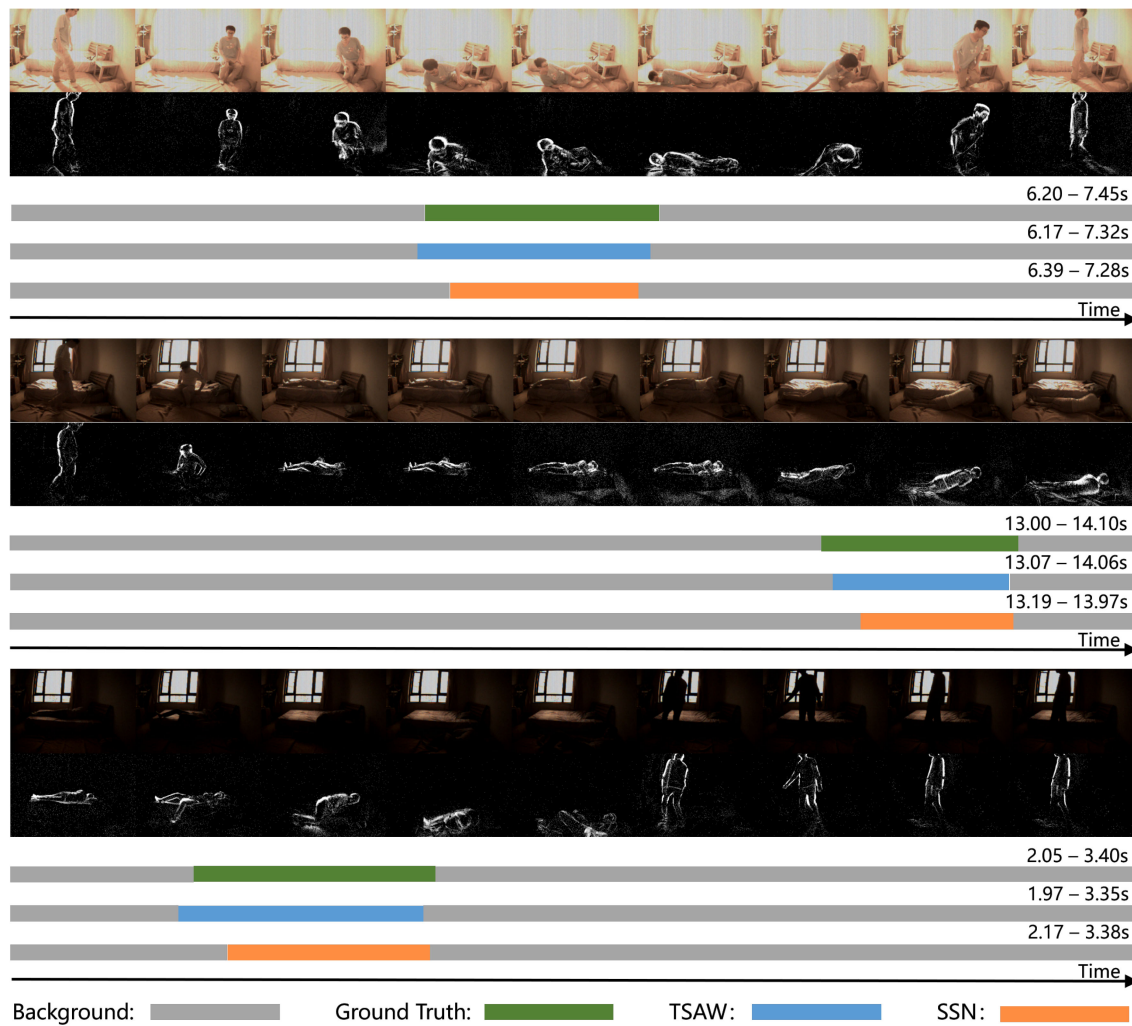


Fig. 9. Qualitative results. We show qualitative falling down action detection results on our event camera-based dataset at different light conditions. We can see that our event camera-based falls temporal detection framework can work properly even under very dark-light conditions.

dataset in Fig. 9. From these results, we can see that even under very dark conditions, our proposed event camera-based falls temporal localization framework can still work normally.

V. CONCLUSION

In this article, we proposed a bioinspired event-camera-based falls temporal localization framework, which is also the first falls temporal localization framework, to the best of our knowledge. For the proposal generation, we proposed the first event density-based action proposal generation scheme, which is simple yet more effective than the existing sliding-window-based schemes. To realize proposal temporal modeling, we introduced a temporal-spatial weighted network to extract more robust motion representations. To achieve a better event feature representation for motion recognition and detection, we introduced an adaptive time window aggregation mechanism, which adaptively handles various fast and slow motions to avoid the transformed event mesh tensor being too blurred or barren for recognition. Extensive experiments on our own collected datasets have shown that our proposed event-camera-based fall temporal localization framework can achieve better

performance. For further research, we will extend our framework to more categories of event-camera-based temporal action detection.

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